

SE Lab-1

PES2UG24CS572

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Blood Bank Inventory & Emergency Donor Matcher

The system should:

- monitor blood-component inventory
- monitor shelf life/expiry
- detect critical shortages
- match eligible donors
- notify compatible donors
- support emergency blood requests

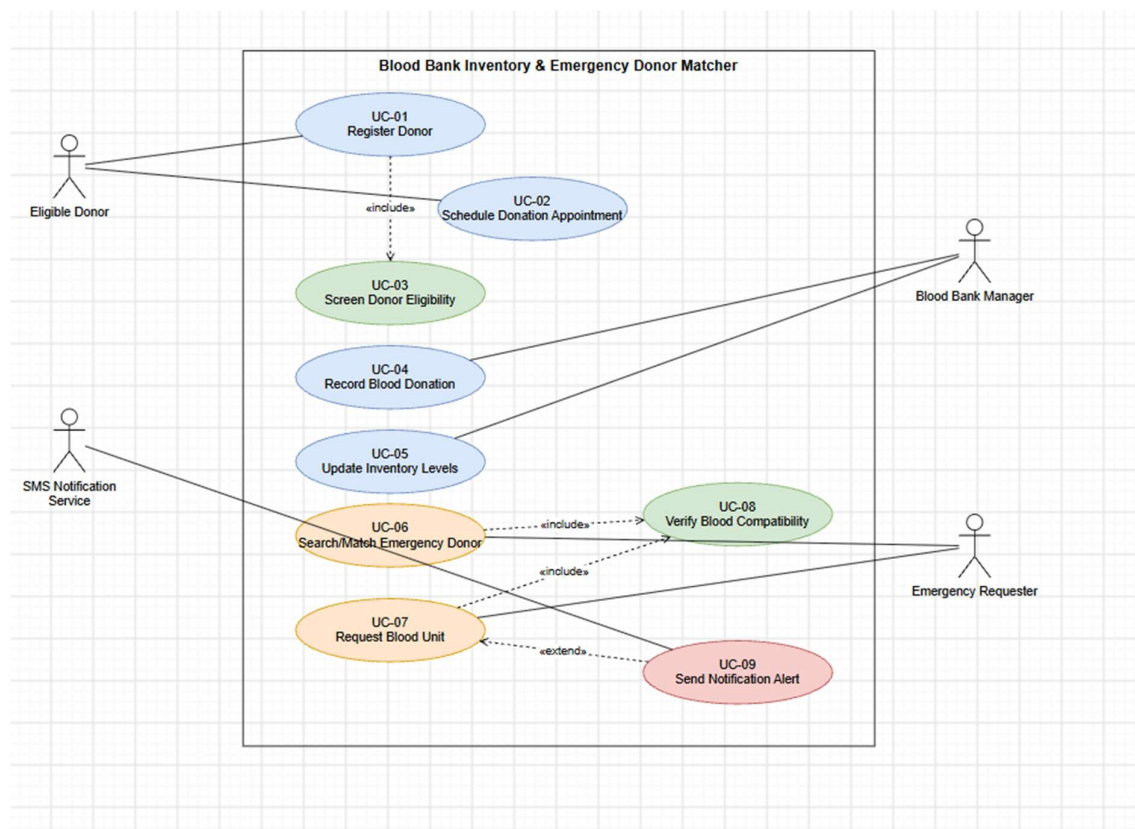
DELIVERABLE 1 — REQUIREMENTS TABLE

Blood Bank Inventory & Emergency Donor Matcher

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	Compatible donors are notified within 30 seconds of an emergency request, and donors with incompatible blood groups are not alerted.	Helps locate suitable blood quickly during critical shortages.
FR-002	Functional	The system shall allow the Blood Bank Manager to view the current inventory of available blood components and their quantities.	High	When the manager requests the inventory, the system displays the current blood group, component type, quantity, and availability status.	Enables the manager to make quick decisions during emergencies.
FR-003	Functional	The system shall monitor the shelf life of stored blood components and identify units approaching their expiry date.	Medium	The system identifies and displays blood units that are within the configured expiry-warning period.	Helps prevent wastage of blood units and supports proper inventory management.
FR-004	Functional	The system shall identify eligible donors based on blood-group compatibility and their registered location.	High	When a blood request is received, the system returns only donors who have a compatible blood group and are within the required search radius.	Ensures that emergency notifications are sent only to suitable donors.

FR-005	Functional	The system shall allow the Blood Bank Manager to update blood inventory when blood units are added, allocated, or removed.	High	After an inventory update, the corresponding blood-unit quantity is correctly reflected in the system.	Keeps inventory information accurate and up to date.
NFR-001	Non-Functional — Performance & Security	The inventory ledger must maintain strict transactional consistency, preventing simultaneous allocation of the same blood bag.	High	Under simulated concurrent allocation requests, a blood unit is allocated to only one request and no duplicate allocation occurs.	Prevents incorrect blood-unit allocation and maintains data consistency.
NFR-002	Non-Functional — Security	The system shall restrict blood inventory modification operations to authenticated Blood Bank Managers.	High	Unauthorized users attempting to modify inventory are denied access in all tested cases.	Protects sensitive inventory information and prevents unauthorized changes.

DELIVERABLE 2 — UML USE-CASE DIAGRAM



DELIVERABLE 3 — USE-CASE FLOW

Use Case: UC-01 — Submit Emergency Blood Request

Primary Actor

Emergency Requester

Supporting Actors

- Blood Bank Manager
- Eligible Donor
- SMS Notification Service

Preconditions

1. The Blood Bank Inventory & Emergency Donor Matcher system is operational.
2. The Emergency Requester is authorized to submit an emergency blood request.
3. Blood inventory and donor information are available in the system.

Main Success Scenario

1. The Emergency Requester submits an emergency blood request by providing the required blood group, blood component, quantity, and emergency details.
2. The system validates the request information.
3. The system records the emergency blood request.
4. The system checks the real-time blood bank inventory for the requested blood component.
5. The system determines whether the required blood is available in sufficient quantity.
6. If additional blood is required, the system identifies eligible donors with a compatible blood group.
7. The system filters compatible donors based on the required location radius.
8. The system sends an emergency SMS notification to the eligible compatible donors.
9. Eligible donors receive the notification and can confirm their donation availability.
10. The system records the donor responses.
11. The system updates the emergency request status with the available blood and donor information.
12. The Blood Bank Manager reviews the request status and available matches.
13. The use case ends successfully.

Alternate Flow 1 — No Compatible Donor Available

7a. No eligible compatible donor is found within the required radius.

1. The system determines that no compatible donor is available within the required search radius.
2. The system displays a message indicating that no immediate donor match is available.
3. The system notifies the Blood Bank Manager about the shortage.
4. The emergency request remains active for further action.
5. The Blood Bank Manager attempts to obtain the required blood through another available source.

Alternate Flow 2 — Required Blood Available in Inventory

5a. The required blood is available in sufficient quantity.

1. The system identifies sufficient compatible blood units in the inventory.
2. The system reserves the required blood units for the emergency request.
3. The system updates the available inventory quantity.
4. The system informs the Emergency Requester that the required blood is available.
5. The use case ends successfully without requiring donor notifications.

Postconditions

1. The emergency blood request is recorded in the system.
2. The blood inventory is updated if blood units are allocated.
3. Eligible donor responses are recorded when donor matching is required.
4. The emergency request status is updated for the Blood Bank Manager.